Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science *West Lafayette, IN*
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics | GPA: 3.32/4.0 May 2028

PROFESSIONAL EXPERIENCE

Boston Dynamics *Waltham, MA*
Sensors and Perception Co-Op – Atlas | Python, Flask, SQL, MuJoCo, CATIA, Linux July 2026 – Present

- Building an impact forensics pipeline and Flask/Plotly dashboard covering ~28k impact events from ~900 robot test logs.
- Running ETL from a time-series telemetry store into SQLite, with a 3D impact-location heatmap and per-event waveform inspection.
- Mapping impacts onto the robot's mesh by propagating 6-DOF floating-base poses through quaternion forward kinematics.
- Estimating impact energy from Khatib's operational-space effective mass, validated against hand-derived Jacobians on a custom arm rig.
- Devising a delta-v severity metric that holds when sensors clip an impact's peak, validated on synthetic pulses.
- Predicting steady-state temperatures across hundreds of units with Savitzky-Golay filtering and multi-start least-squares fits.
- Designing CATIA fixtures and shear features, using pin-shear failure analysis to optimize geometry.

Caterpillar (via Purdue Data Mine) *West Lafayette, IN*
Machine Learning Researcher | Python, Snowflake, Power BI January 2026 – May 2026

- Built LSTM and XGBoost models with Monte Carlo confidence intervals forecasting paint and solvent use, cutting material usage 4.7%.
- Consolidated five years of noisy legacy files into one daily series surfaced through Power BI, cutting cycle time 3.3%.

Chortos Lab – Soft Adaptive Systems and Structures *West Lafayette, IN*
Undergraduate Researcher | Python, LabVIEW, MATLAB, Fusion 360 August 2025 – May 2026

- Built LabVIEW and MATLAB acquisition reading a Keyence laser displacement head through an NI USB-6001.
- Wrote a Python computer-vision tracker resolving cantilever displacement to 0.01 mm across 50,000+ point datasets.
- Formulated ionic dielectric gel variations, tuning chemical composition to raise electromechanical force output over 50%.

Emerson – ASCO *Florham Park, NJ*
Mechanical & Manufacturing Engineering Intern | Creo Parametric, CNC, GD&T May 2025 – August 2025

- Engineered 30+ rotary valve components in Creo Parametric, validated with hand-calculated stress analysis.
- Machined precision pistons to ±0.05 mm, authoring an assembly procedure that cut actuator deadband over 10%.

PROJECTS

Purdue ASME – Biomechanics *West Lafayette, IN*
Active Orthosis – Electronics Team Lead | C++17, ESP32-S3, FreeRTOS, KiCad, GoogleTest April 2024 – May 2026

- Designed a 1 kHz PI velocity controller in C++17 on an ESP32-S3 with feed-forward and anti-windup.
- Cut EMG-to-motion latency from 103 ms to 43 ms with a Bayesian estimator, validated on 24 real contractions.
- Protected the motor with 10 latching fault monitors, a hardware watchdog, and a kill switch.
- Led a 5-engineer team redesigning the 4-channel EMG board in KiCad with an INA333 front end and ISO7763 isolation.
- Raised day-to-day gesture decoding from 31% to 87% in emulated tests.

Asteroid Hopping Robot – Independent Research *West Lafayette, IN*
Purdue University & Universidad Carlos III de Madrid | MuJoCo October 2026 – Present

- Building a three-legged hopping robot with reconfigurable leg geometry, plus a pendulum rig to test landings.
- Designing push-off control for target launch velocity, knee impedance control for touchdown, and reaction-wheel attitude control.
- Sizing hops in MuJoCo, comparing risk-limited policies that trade distance against escape velocity and tipover.

Legged Robot State Estimation and Control *Personal Project*
Contact-aided EKF | C++/Eigen, MuJoCo, GoogleTest, Python August 2026 – October 2026

- Built a contact-aided error-state EKF in C++/Eigen fusing a 1 kHz IMU with leg kinematics on a MuJoCo quadruped.
- Held base velocity to 0.013 m/s RMSE over a one-minute walk, against 1.38 m/s for IMU-only dead reckoning.
- Verified every Jacobian against central finite differences to 1e-8 and matched C++ to its Python reference across 55 tests.
- Closed the loop driving Raibert foot placement, attitude PD, and QP balance from the estimate: 5 seeds, 0 falls.

SKILLS & INTERESTS

Programming & Tools: C++17, Python, MATLAB, SQL, LabVIEW, Eigen, MuJoCo, FreeRTOS, GoogleTest, Git, Linux, KiCad, CAD (CATIA, Creo, NX), FEA

Methods & Systems: Sensor Fusion, State Estimation, Kalman Filtering, Real-Time Embedded Control, Signal Processing, Computer Vision, ETL

Languages: English (Native), Turkish (Fluent), Spanish (Fluent – NJ Seal of Biliteracy) | **Hobbies:** Lifting, Hiking, Guitar, Saxophone